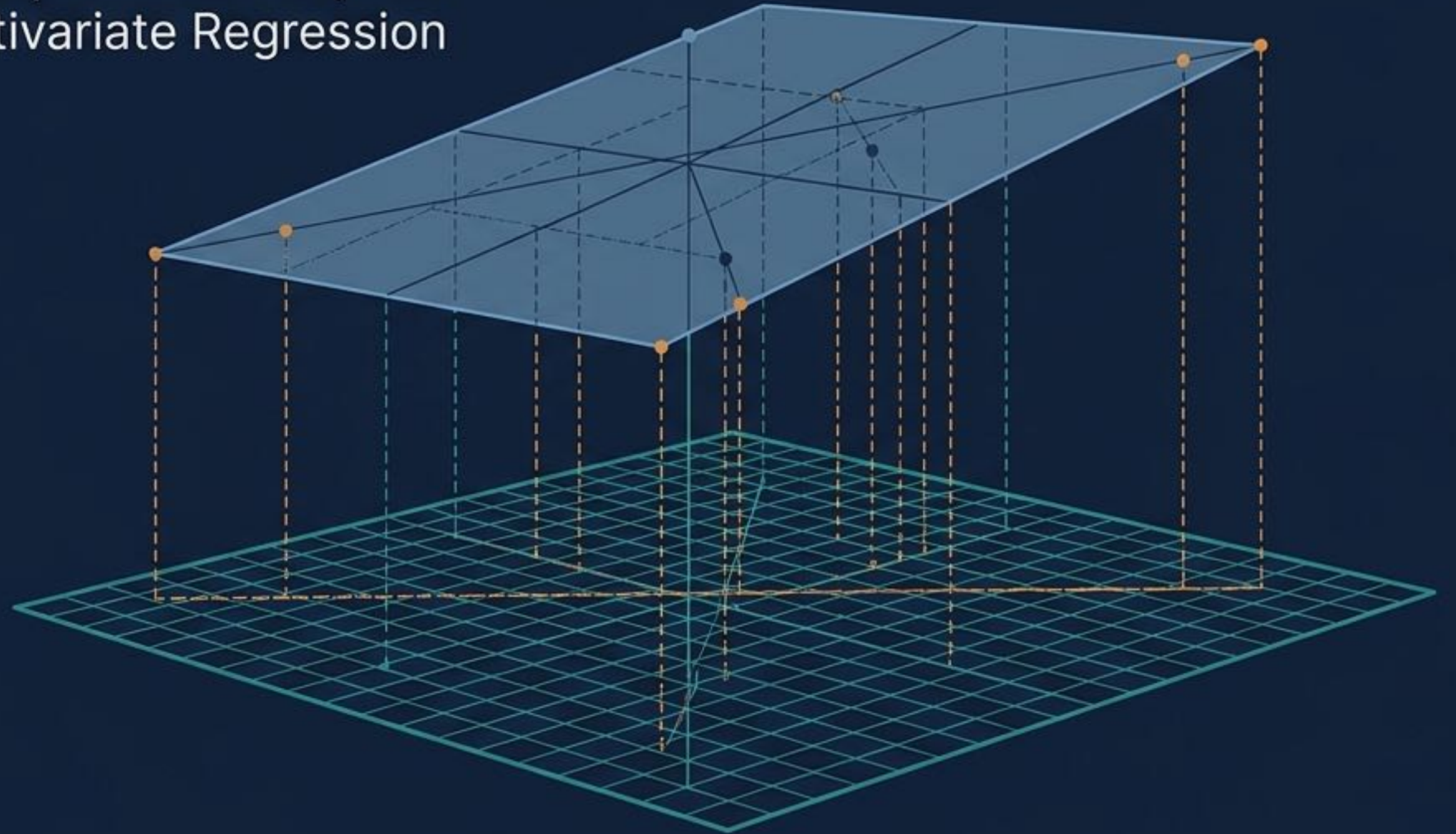


The Econometrician's Blueprint: Unpacking the Frisch-Waugh-Lovell Theorem

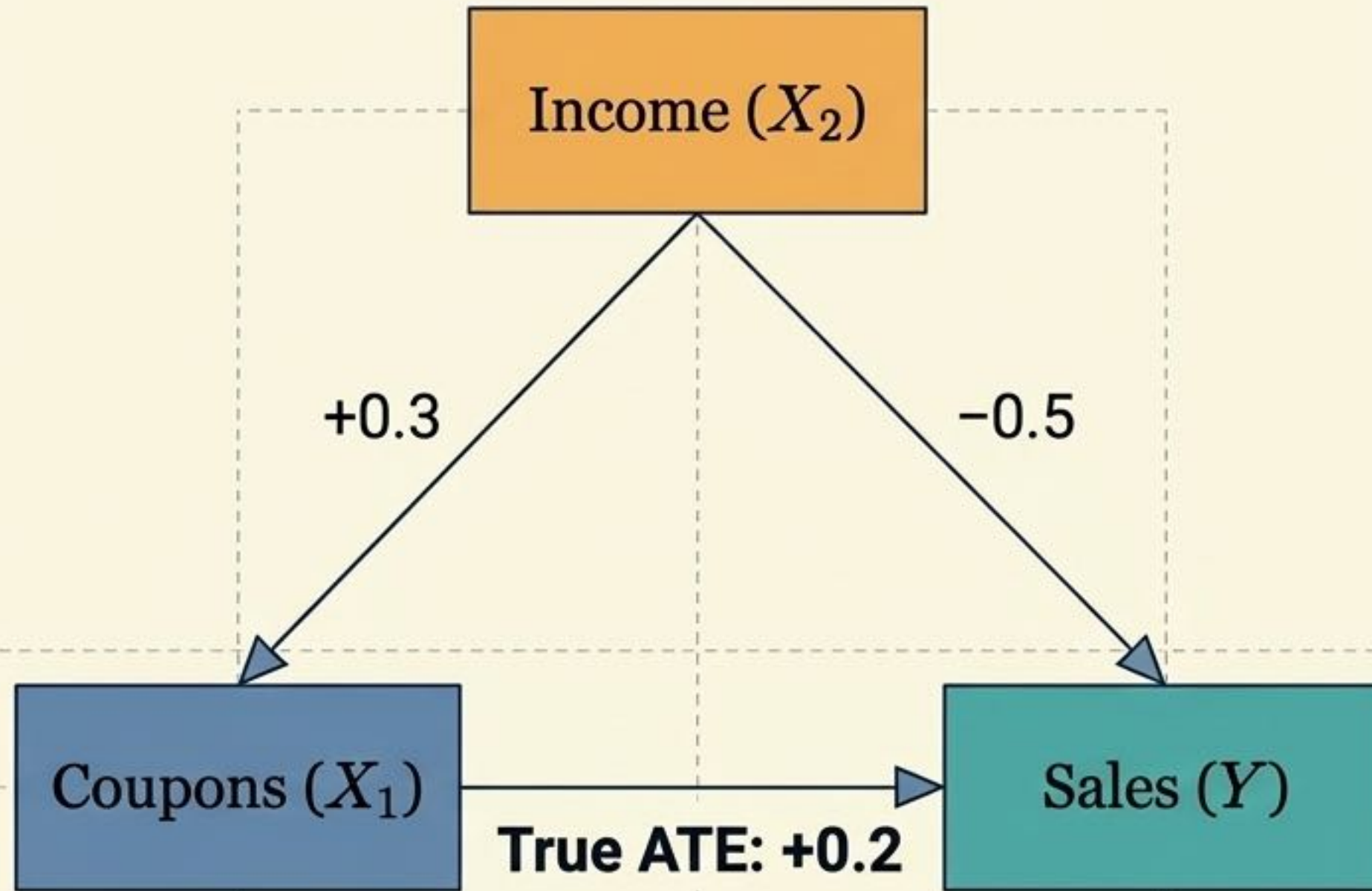
A Matrix Proof, Algorithmic Implementation, and
Intuitive Visualization of Multivariate Regression



Core Lexicon and Intuitive Analogies

Concept	Formal Definition	Intuitive Analogy
Partialling-Out (Residualization)	$\tilde{y} = y - \hat{y}$ Isolating the variation orthogonal to the controls	 Judging a runner's time by explicitly subtracting the measured effect of a tailwind.
Confounding	$X_1 \leftarrow X_2 \rightarrow Y$ A backdoor path creating non-causal association	 Hot weather (confounder) driving simultaneous spikes in both ice cream sales and drownings
Marginal vs. Conditional Effect	$E[Y \mid X_1] \text{ versus } E[Y \mid X_1, X_2]$ Comparing housing prices city-wide (marginal) versus strictly within a single neighborhood	 Comparing housing prices city-wide (marginal) versus strictly within a single neighborhood (conditional)

The Causal DGP: Confounding in Retail Store Performance



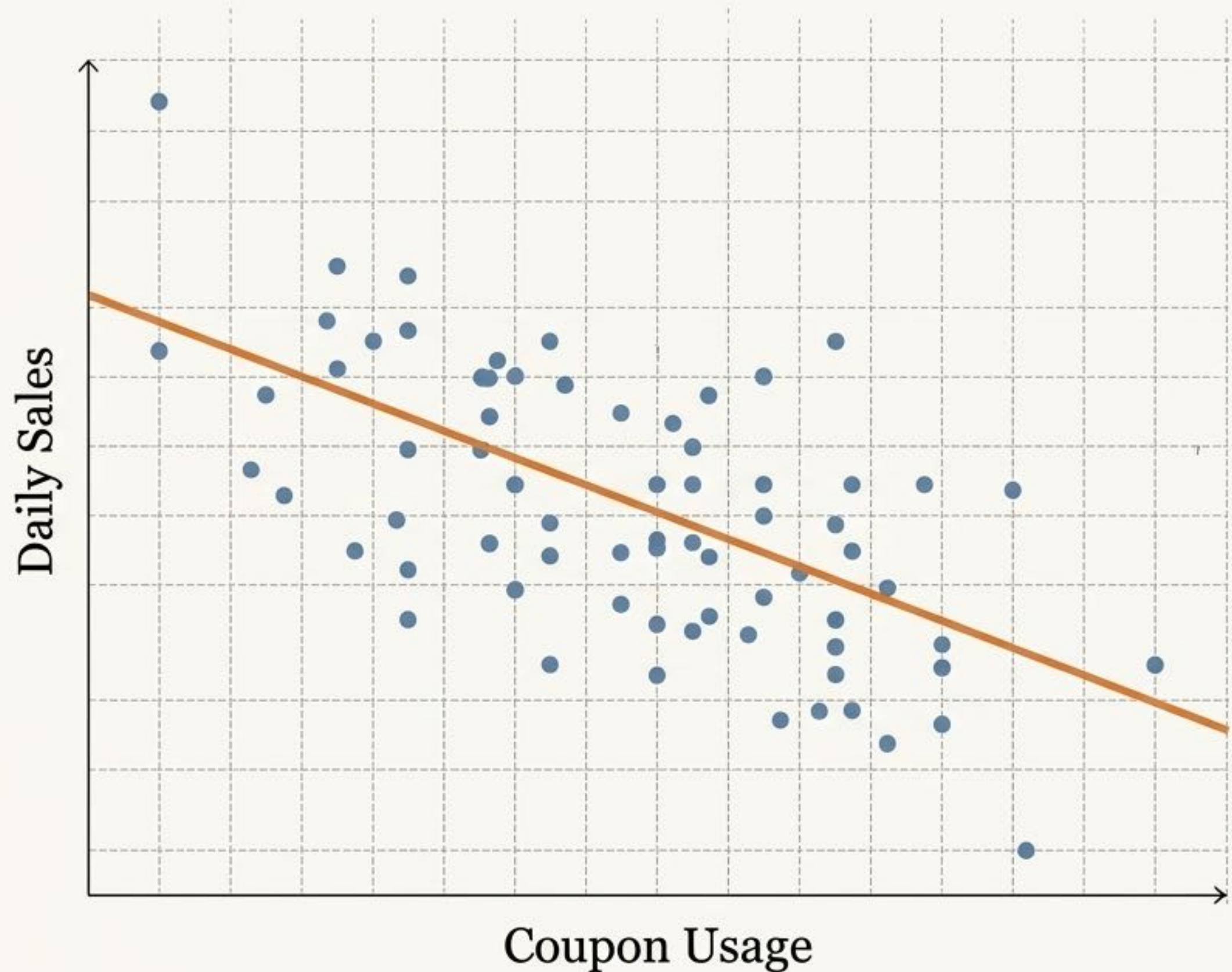
$N = 50$ stores.

$\text{income} \sim N(50, 10)$

True Causal Effect of Coupons on Sales: Exactly $+0.2$ (an average increase of \$200 per 1% coupon increase).

Key Insight: Wealthier neighborhoods use fewer coupons but inherently spend more, creating a backdoor path that masks the true positive effect.

Simpson's Paradox and the Naive Regression



```
smf.ols("sales ~ coupons", df)
```

Naive Slope:	-0.1059
p-value:	0.365

A rushed analyst sees the negative slope and concludes coupons hurt sales. But the true ATE is +0.2. +0.2. Confounding from income has dragged the slope entirely in the wrong direction, masking the true effect.

Decomposing the Gap: The Omitted-Variable Bias Identity

$$\text{beta_naive} = \text{beta_true} + \text{gamma} * \text{delta}$$

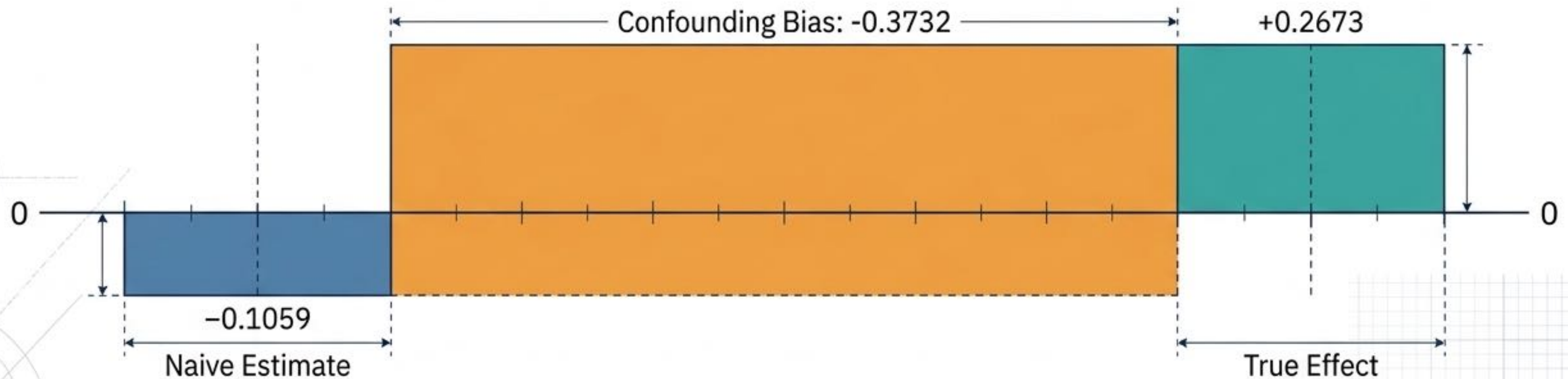
beta_naive (Naive Slope) = -0.1059

gamma (Income's effect on Sales) = +0.3836

beta_true (True Conditional Effect) = +0.2673

delta (Income's slope on Coupons) = -0.9730

$$\text{Calculation: } -0.1059 = 0.2673 + (0.3836 \times -0.9730)$$



Key Insight: The bias is exactly the omitted variable's effect on the outcome multiplied by the slope of the omitted variable on the treatment.

The FWL Theorem: Formal Statement and Matrix Proof

The Statement

For the model $y = X_1\beta_1 + X_2\beta_2 + \epsilon$, the OLS estimator β_1 is identical to:

$$\beta_{1,\text{FWL}} = (\tilde{X}_1^T \cdot \tilde{X}_1)^{-1} \cdot \tilde{X}_1^T \cdot \tilde{y}$$

$\tilde{X}_1$ and $\tilde{y}$ are the residuals from regressing X_1 and y on X_2 .

The Proof

Define the residual-maker matrix:

$$M_2 = I_n - X_2 \cdot (X_2^T \cdot X_2)^{-1} \cdot X_2^T$$

Properties: $M_2 \cdot X_2 = 0$ (Annihilates controls), M_2 is symmetric and idempotent.

Line 1: $y = X_1\beta_1 + X_2 \cdot \beta_2 + e$

Line 2 (Premultiply by M_2):

$$M_2 \cdot y = M_2 \cdot X_1\beta_1 + e$$

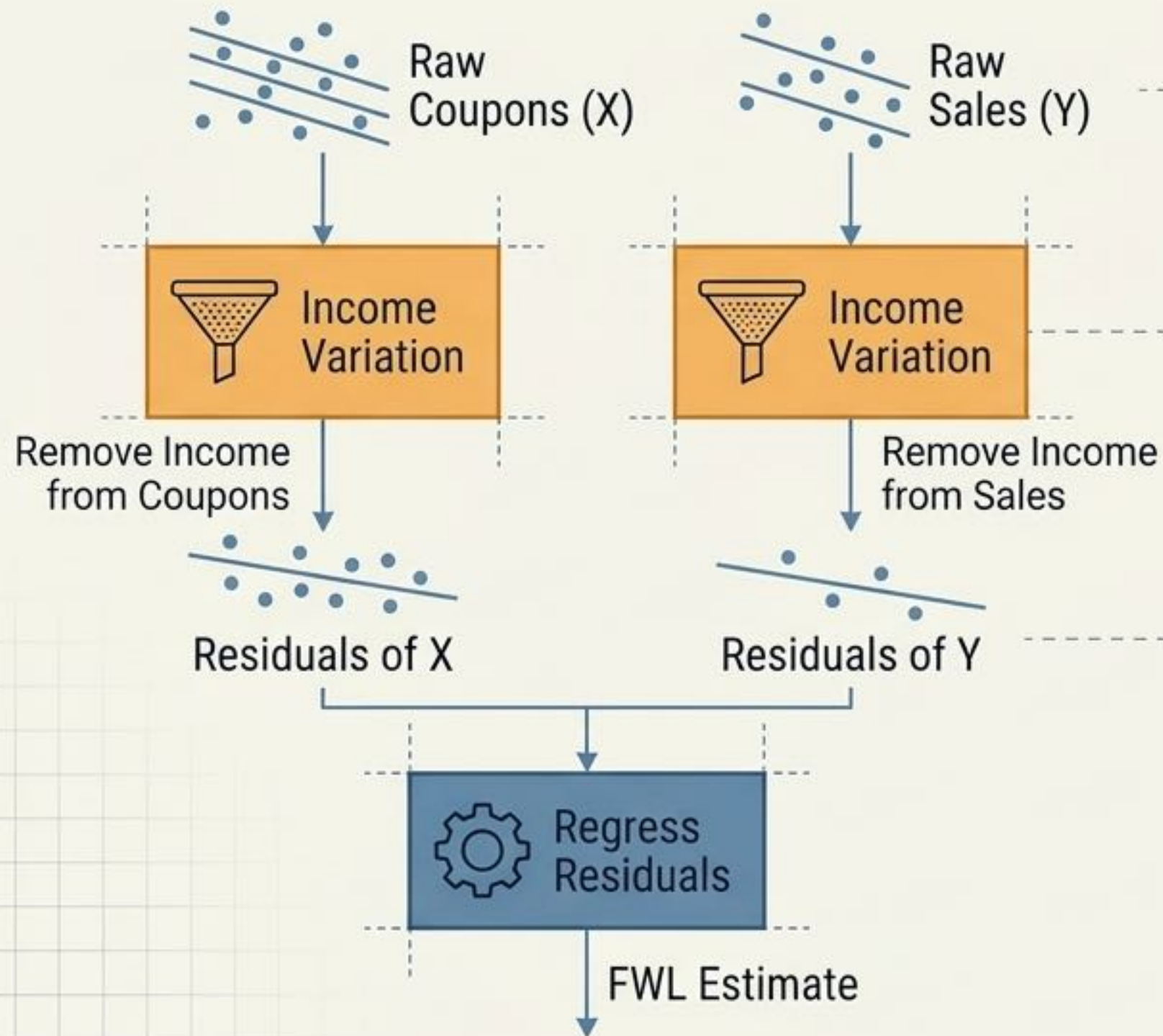
Line 3 (Premultiply by X_1^T):

$$X_1^T \cdot M_2 \cdot y = X_1^T \cdot M_2 \cdot X_1\beta_1$$

Line 4 (Solve):

$$\beta_1 = (\tilde{X}_1^T \cdot \tilde{X}_1)^{-1} \cdot \tilde{X}_1^T \cdot \tilde{y}$$

Algorithmic Execution: Implementing FWL in Software



```
coupons_tilde = ols('coupons ~ income').resid
```

```
sales_tilde = ols('sales ~ income').resid
```

```
ols('sales_tilde ~ coupons_tilde - 1').fit()
```

```
Result: beta_1 = 0.2673 (Identical to the  
full multivariate model)
```

Inferential Nuances:

Thenferential Nuances: The Anatomy of Standard Errors

Method	Coefficient	Standard Error	Residual DF	Sum of Squared Residuals
Full Model	0.2673	0.1203	47	490.8
Step 1 (Residualize X only, No Intercept)	0.2673	1.2715	49	57,181.2
Step 2 (Residualize both)	0.2673	0.1178	49	490.8

Why does Step 1 explode?

Missing the intercept forces the model to explain the baseline level of sales, exploding the SSR. Adding the intercept closes 98% of the SE gap.

Why does Step 2 slightly underestimate SE?

The software doesn't know two parameters (intercept, income) were pre-estimated. The Step 2 SE (0.1178) must be multiplied by $\sqrt{49/47}$ to perfectly match the Full Model (0.1203).

Under the Hood: FWL via Matrix Algebra in NumPy

$$\beta_{1FWL} = \frac{\text{Cov}(\tilde{y}, \tilde{x}_1)}{\text{Var}(\tilde{x}_1)}$$

Calculating FWL without regression libraries relies directly on extracting orthogonalized vectors using least squares, then computing sample covariance.

```
c_tilde = x1 - X2 @ np.linalg.lstsq(X2, x1)[0]
s_tilde = y - X2 @ np.linalg.lstsq(X2, y)[0]
```

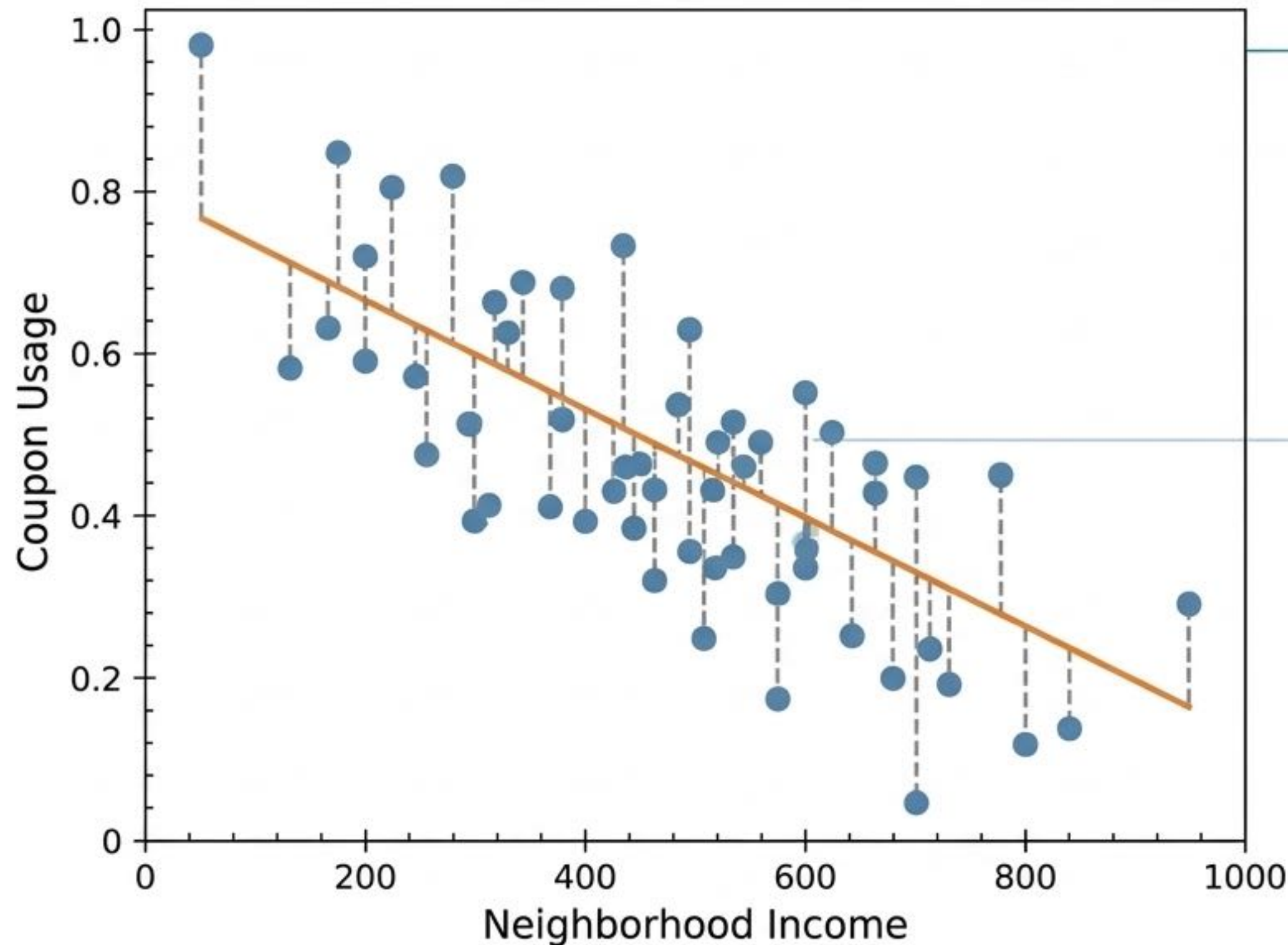
Line 5 of the matrix proof

```
cov_sc = np.cov(s_tilde, c_tilde)[0, 1]
var_c = np.var(c_tilde, ddof=1)
beta_1 = cov_sc / var_c
```

Outputs 0.2673

Caution: `np.cov` divides by $n-1$. `np.var` divides by n unless `ddof=1` is passed. Mixing divisors artificially inflates the slope to 0.2728.

Visualizing Partialling-Out: Isolating Orthogonal Variation



The Mechanical Geometry of OLS

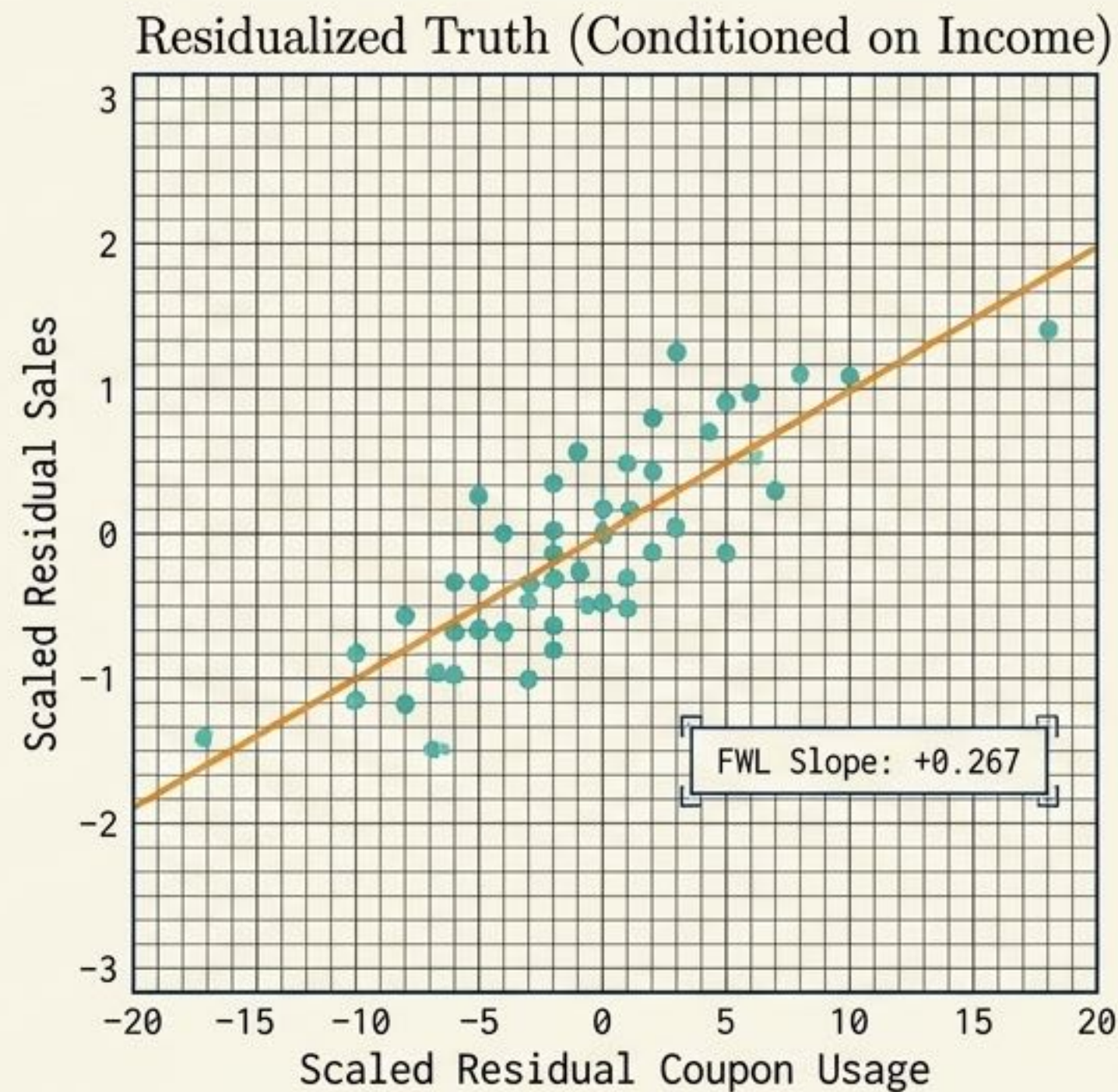
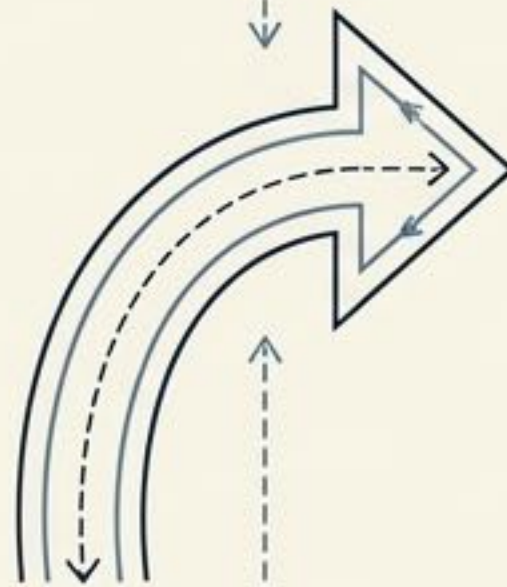
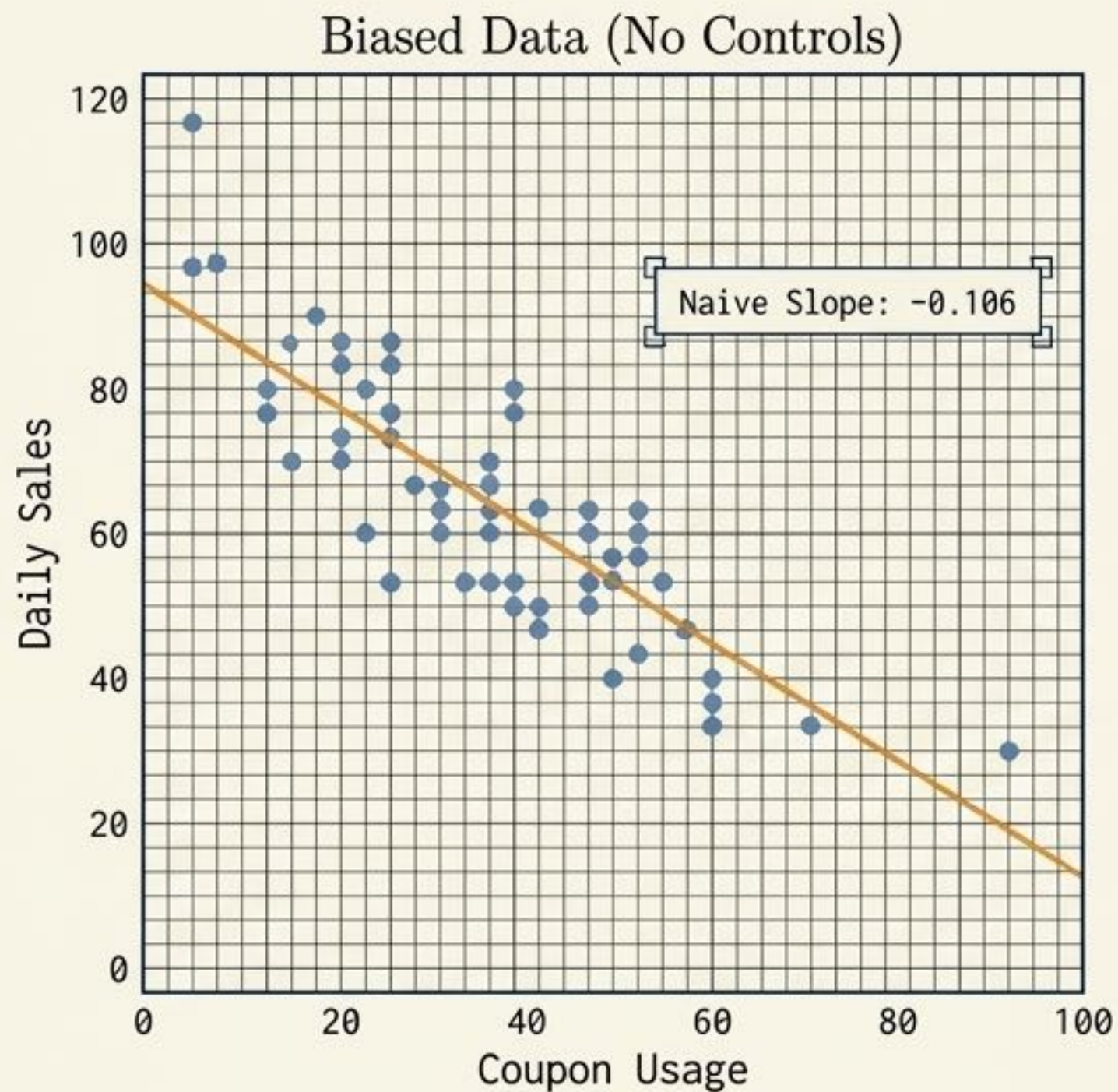
The fitted warm orange line represents expected coupon usage given income.

The Residuals ($x_1\text{-tilde}$)

The vertical dashed lines represent the variation FWL actually uses. FWL discards the fitted line entirely and only analyzes these vertical distances.

Synthesis: “Among stores with the exact same neighborhood income, which ones have unusually high or low coupon usage?”

Simpson's Paradox Resolved: The Conditional Relationship



Technical Note: Once the confounding filter of income is applied, the true causal direction emerges. Scaling the residuals by adding back the sample mean restores interpretable, real-world axis units without altering the 0.2673 slope.

Generalizing FWL: Multiple Controls and Fixed Effects

Multiple Controls

- FWL seamlessly handles an N-dimensional matrix of controls simultaneously.
- Adding 'dayofweek' to the model shifts the coupon coefficient slightly from 0.2673 to 0.2706.
- Insight: This shift is driven entirely by the partial correlation (-0.021) between coupons and day-of-week after holding income fixed.

```
ols('sales ~ coupons + income + dayofweek')
```

Fixed Effects Connection

How do high-dimensional FE packages (e.g., `reghdfe` in Stata, `pyfixest` in Python) handle models with 10,000+ dummy variables without crashing?



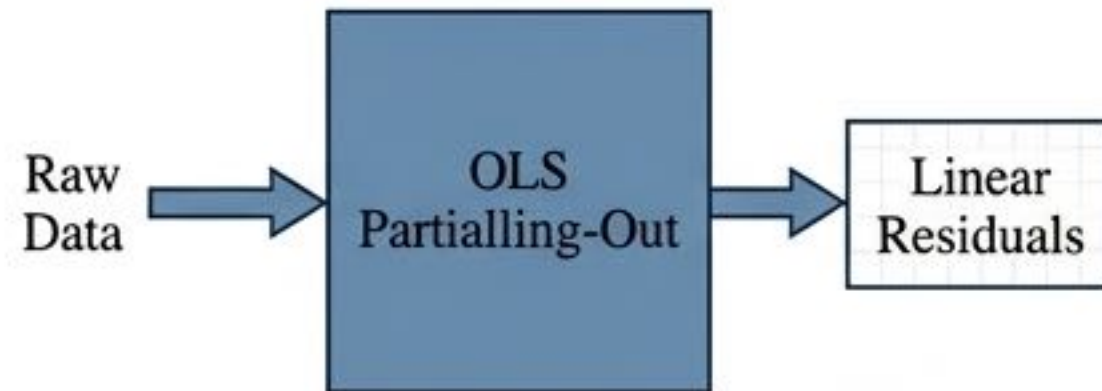
They use the FWL theorem. They residualize the data by demeaning within groups (subtracting group means) to completely bypass the impossible computational task of inverting massive fixed-effect matrices.

Methodological Synthesis: From FWL to Double Machine Learning (DML)

The Limitation:

Linear partialling-out fails if the outcome depends on the controls non-linearly (e.g., Income^2).

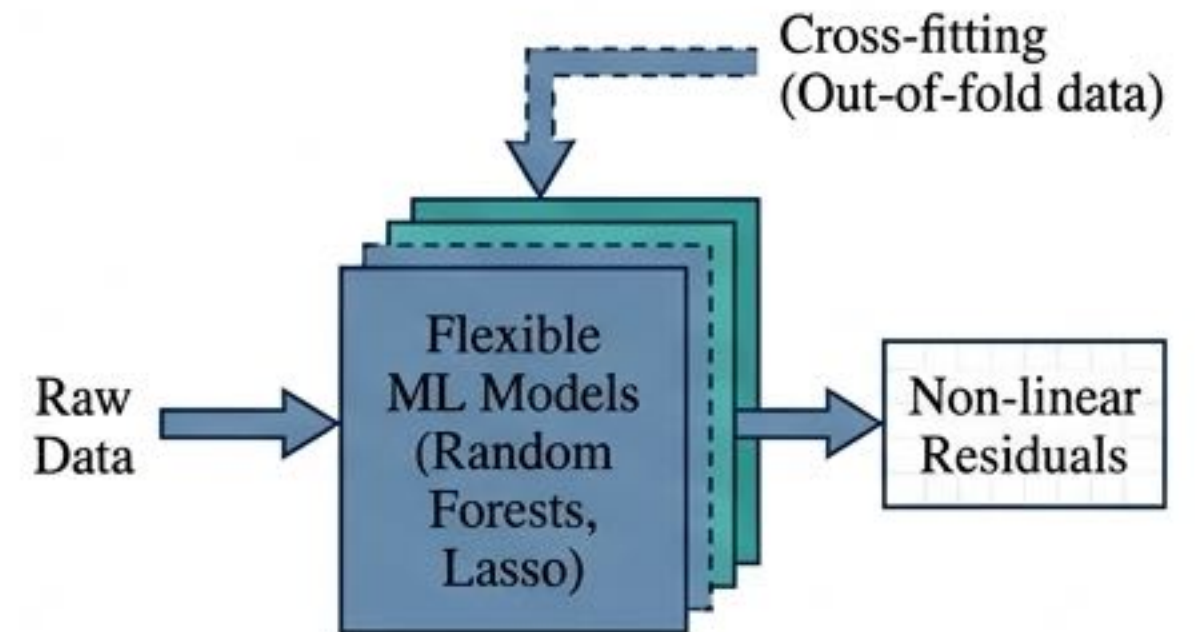
Linear FWL



The DML Innovation:

- Retains the exact mathematical FWL logic (residualize treatment, residualize outcome, regress residuals).
- Swaps the rigid OLS models for flexible Machine Learning estimators.
- Adds cross-fitting to prevent models from memorizing a store's own noise into its residuals.

Modern DML



Diagnostic Checks: Common Methodological Misconceptions

The Myth	The Mathematical Truth
Myth: FWL only works for a single control variable.	Truth: FWL handles an N-dimensional matrix of controls simultaneously; it is the basis of modern Fixed Effects software.
Myth: Uncorrelated OLS residuals means the residuals are statistically independent of controls.	Truth: Linear OLS only guarantees zero linear correlation. Residuals can still have massive dependency on non-linear transformations (e.g., Income^2).
Myth: Because FWL isolates orthogonal variation, it automatically guarantees a causal effect.	Truth: FWL is just matrix algebra. Causal identification still strictly requires the theoretical DAG assumption of no unmeasured confounding.

The Estimator Synthesis: Summary Reference Table

Method	Coefficient	Std. Error	p-value	Residual DF
Naive OLS	-0.1059	0.1158	0.365	48
Full OLS (+ income)	+0.2673	0.1203	0.031	47
FWL Step 1 (no intercept)	+0.2673	1.2715	0.834	49
FWL Step 1 (+ intercept)	+0.2673	0.1437	0.069	48
FWL Step 2 (residualize both)	+0.2673	0.1178	0.028	49
FWL (+ income + day)	+0.2706	0.1157	0.023	49

Ultimate Diagnostic Rule: The coefficient survives every FWL shortcut; the standard error does not. Always correct degrees of freedom or report the full-model SE.